SUPERVISOR CORRECTIONS AND PANEL DEMONSTRATION GUIDE

Companion to the revised Chapter 1. This guide records the response to the 12 supervisor comments and explains how the objectives can be demonstrated. It is not an additional dissertation chapter or evidence that the proposed acceptance tests have all passed. The original document and application were left unchanged.

1. Supervisor Corrections

Comment	Location	Correction
0 / Introduction	1.1	Rewritten opening; concise purpose and chapter map; removed feature counts, dataset composition, model selection, threshold and repeated scope statements.
1 / Background	1.2	Added named studies with methods, findings, limits and relevance; used an international-to-Zimbabwe progression; stated a bounded integration gap and removed project implementation detail.
2 / Problem	1.3	One short problem paragraph; transactions terminology; difficulty, monitoring limitation and consequence; removed the proposed solution.
3 / Aim	1.4	One development verb and one sentence linking behavioural analysis, machine learning and investigation support.
4 / Transaction objective	1.5, Objective 1	Week 3 deadline, three operation types, 12 acceptance cases, balance reconciliation and duplicate-request protection.
5 / Behaviour objective	1.5, Objective 2	Week 6 deadline, six histories across three behavioural dimensions, independently calculated expected values and chronological correctness.
6 / Investigation objective	1.5, Objective 4	Week 10 deadline and six explicit checks for alert provenance, review, recorded status, access and notifications.

Comment	Location	Correction
7 / Limitations	1.6	Explained synthetic-data necessity and representativeness; limited generalisation; acknowledged model errors; described actual local timing scope without inventing resource restrictions.
8 / Delimitations	1.7	Combined deployment exclusions; complete sentences; maintained internal reports, rules, investigations and supported transaction types.
9 / Versions	1.8	Added inspected hardware and software versions; distinguished application and model-runtime environments.
10 / Justification	1.10	Focused on local value, investigator needs and an integrated academic contribution; removed implementation details and the unsupported reference to real data.
11 / Summary	1.11	Short summary of problem, objectives, scope and contribution; explicit transition to Chapter 2.

Additional guideline corrections: five process-based objectives beginning with “To”; dashboard construction replaced by reliable monitoring as the fifth objective; Times New Roman 12-point body text, 14-point chapter heading, numbered headings, 1.5 line spacing, numbered tables and a 12-week Gantt chart. IEEE-style chapter-local references correct the Zimbabwe study’s authors to Chitimira and Torerai. When combining the dissertation, merge these entries into the master bibliography and renumber citations consistently across all chapters.

2. Demonstrating the Five Objectives

Use prepared fixtures and evidence from a dedicated demonstration environment. Do not reset the working database or execute live payments merely to complete this guide. The durations below are presentation allocations, not measured algorithm runtimes. Financial demonstrations require the agreed test authorisation and before/after reconciliation. Candidate-model comparison can use the retained experiment records without retraining.

2.1 Objective 1 / Week 3 / 4 minutes

Demonstration: Show the 12-case transaction matrix: four cases per operation (valid, invalid, insufficient funds and repeated request). Display expected and actual wallet/agent ledger effects.

Acceptance: All 12 cases match the expected result. Repeated request IDs produce one financial effect. Cash operations require configured demo-agent balances or isolated fixtures.

2.2 Objective 2 / Week 6 / 4 minutes

Demonstration: Trace six saved histories, one routine and one suspicious-pattern example per behavioural family. Show source records, calculation cut-off and independently calculated expected feature values.

Acceptance: All six calculation comparisons agree within documented numerical tolerance. Later records do not enter earlier feature calculations. This is feature correctness, not a promise that all suspicious examples are detected.

2.3 Objective 3 / Week 9 / 5 minutes

Demonstration: Open the four-candidate results and timing records. Explain the selection trade-off. Show saved normal and suspicious outputs and their corresponding stored decisions.

Acceptance: Complete, comparable metrics for four candidates; traceable model identity and exact decision agreement. Missing or failed inference is identified explicitly. Report misclassifications honestly.

2.4 Objective 4 / Week 10 / 4 minutes

Demonstration: Show AI-only, rule-only and combined alerts; review and record a case status; attempt customer access; verify staff-only notification delivery.

Acceptance: All six cases pass. Rule findings remain separate from AI findings, and customer access to compliance data/events is denied.

2.5 Objective 5 / Week 12 / 4 minutes

Demonstration: Present isolated tests for fresh history, concurrent change, inference failure, audited retry and duplicate payment requests. Display transaction IDs, audit entries and balances.

Acceptance: All five cases preserve financial integrity. Verification or inference failure does not replay a payment, fabricate a prediction or silently select the legacy model.

The 12/6/6/5 case counts are explicit acceptance criteria introduced in this revision. Existing tests should be mapped to these cases; any uncovered case must be marked pending. This document revision did not execute or certify the application tests. Chapters 3 and 4 should contain the case IDs, expected outcomes, actual outcomes and supporting evidence.

3. SMART Commitments for Each Learning Algorithm

Specific: evaluate each named candidate for the same suspicious-pattern classification task. Measurable: record the confusion matrix, precision, recall, macro-F1, false-positive rate and training time. Achievable: retained results already exist for all four candidates. Relevant: compare detection errors and computational cost to support the primary-model decision. Time-bound: complete the individual milestones in Weeks 7–9 and the shared comparison by the end of Week 9. These are the retained schedule's milestones, not a claim that experiment timestamps independently prove the original development start date.

Table 3.1: Retained candidate training times and development-validation results

Candidate	Deadline	Fit time (s)	Precision / recall
Logistic Regression	Week 7	0.989	0.8092 / 0.5977
Random Forest	Week 8	20.998	0.9462 / 0.8825
Histogram-based Gradient Boosting	Week 8	4.903	0.9527 / 0.9248
XGBoost	Week 9	2.632	0.9587 / 0.9036

Source: ml_v4/evidence/stage18/resource_runtime_report.json and comparison.json, primary candidate runs. Recorded configuration: one thread, 60,000 development-training rows and 15,000 development-validation rows. Times measure estimator fitting in that experiment, not feature construction, application start-up or payment-to-alert latency. They are observations, not guaranteed maximum runtimes. The full metrics and confusion matrices are retained in comparison.json.

For the panel, allocate one minute per algorithm to show its saved metrics and observed fit time, then one minute to explain selection. A fresh training run is unnecessary. If end-to-end inference speed is claimed, separately measure elapsed time from committed transaction to

persisted inference outcome on the stated hardware and history size. No such latency measurement was made during this document revision.

4. Alignment with the Current V4 System

The current frozen authority and retained dissertation evidence identify XGBoost as the primary V4 model, with 40 semantic features, 42 estimator input columns and a 0.50 decision threshold. These implementation facts belong in the methodology and results chapters. The revised introduction and background no longer repeat the old 30-feature architecture, Gradient Boosting selection or 0.35 threshold. The four compared algorithms were Logistic Regression, Random Forest, Histogram-based Gradient Boosting and XGBoost.

The retained XGBoost development-validation precision was 0.9587 and recall was 0.9036. These figures are synthetic development-validation results, not independent-test or real-world performance. Histogram-based Gradient Boosting had higher recall and macro-F1 in the same comparison, whereas XGBoost had a lower false-positive rate. Explain the documented selection trade-off; do not claim that XGBoost was best on every measure. The retained summary also records uneven detection across scenario families, so overall metrics must not be presented as proof of uniform detection.

The evidence summary states that the original final-test and independent-evaluation partitions were quarantined. Do not describe their results as successful independent validation. Do not confuse the generator's routine/suspicious scenario proportions with model accuracy or require predictions to follow the generated class ratio. Independent rule alerts and V4 classifications should remain distinct in the presentation.

Evidence inspected: ml_v4/evidence/stage23/dissertation_results_summary.md; ml_v4/frozen_inference/authority.json; and the two Stage18 files cited above. These were read, not modified. Chapters 2–5 were not rewritten as part of this Chapter 1 correction; align their model descriptions, evaluation labels, objectives and bibliography before combining the final dissertation.